CLASS ASSESMENT QUESTIONS (1-7)

1. A and B started a business with Rs. 1500 and Rs. 2500 and got a profit of Rs. 800. Half of the profit is shared equally the remaining is shared according to their investment. Find their profits respectively.

2. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

3. A shopkeeper sells one transistor for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. His total gain or loss percent is?

4. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is?

5. A product's price was first increased by 20%, and then later decreased by 10%. What is the net percentage change in the product's price?

6. If an article is sold for ₹ 178 at a loss of 11%, what should be its selling price in order to earn a profit of 11%?

7. A dealer uses a scale of 90 cm instead of a metre scale and claims to sell at a cost price. What is his gain%?

8. If Sohan, while selling two goats at the same price, makes a profit of 10% on one goat and suffers a loss of 10% on the other [CSAT 2014]
    1. He makes no profit and no loss.
    2. He makes a profit of 1%.
    3. He suffers a loss of 1%.
    4. He suffers a loss of 2%.

9. A person buys three articles P, Q and R for Rs. 3,330. If P costs 25% more than R and R costs 20% more than Q, then what is the cost of P?

10. A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is?

11. A cyclist rides at an average speed of 20 km/h. If they cover a distance of 50 km, how long did their ride take in hours and minutes?

12. A train travels 300 km in 4 hours. What is its speed? If it maintains this speed, how far will it travel in 7 hours?

13. An airplane flies at a speed of 800 km/h. How long will it take to cover a distance of 2400 km? Express your answer in hours and minutes.

14. Car A leaves City X traveling towards City Y at 70 km/h. At the same time, Car B leaves City Y traveling towards City X at 80 km/h. The distance between City X and City Y is 450 km. How long will it take for the two cars to meet?

15. A person drives to a destination at 50 km/h and returns by the same route at 60 km/h. If the total time for the round trip is 11 hours, what is the one-way distance to the destination?

16. A bus leaves a station at 6:00 AM traveling at 40 km/h. A car leaves the same station at 7:00 AM traveling in the same direction at 60 km/h. At what time will the car overtake the bus?

17. A train travels the first 100 km of its journey at 50 km/h, the next 150 km at 75 km/h, and the final 50 km at 25 km/h. What is the average speed of the train for the entire journey? (Hint: Average speed is total distance / total time).

18. Two runners are on a circular track of 400 meters. They start at the same point and run in opposite directions. Runner 1 runs at 5 m/s and Runner 2 runs at 3 m/s. How long will it take them to meet for the first time?

19. A bus is moving 30km/h ahead of a car moving at 50 km/h. How many km apart are they if it takes 15 min for the car to catch the bus?

20. In a 500 metres race, B starts 45 metres ahead of A, but A wins the race while B is still 35 metres behind. What is the ratio of the speeds of A to B assuming that both start at the same time?

21. A car travels from a place X to place Y at an average speed of V km/hr, from Y to X at an average speed of 2v km/hr, again from X to Y at an average speed of 3v km/hr and again from Y to X at an average speed of 4v km/hr. Then the average speed of the car for the entire journey
    a.is less than v km/hr
    b. lies between v and 2v km/hr
    c. lies between 2v and 3v km/hr
    d. lies between 3v and 4v km/hr

22. If a: b = 5: 9 and b: c = 7: 4, then find a: b: c.

23. Divide Rs. 981 in the ratio 5: 4

24. A bag contains 50 p, 25 p, and 10 p coins in the ratio 2: 5 : 3, amounting to Rs. 510. Find the number of coins of each type.

25. Mixture contains sugar solution and coloured water in the ratio of 4 : 3. If 10 litters of coloured water is added to the mixture, the ratio becomes 4: 5. Find the initial quantity of sugar solution in the given mixture.

26. In a school, the ratio of boys to girls is 7:5. If 20% of the boys and 30% of the girls receive scholarships, and there are 420 boys in the school, find the total number of students who receive scholarships.

27. A 150-meter-long train is running at a speed of 60 km/h. In what time will it pass another train 100 meters long running in the same direction at a speed of 45 km/h?

28. Two trains, one 120 meters long and the other 180 meters long, are moving towards each other on parallel tracks at speeds of 50 km/h and 40 km/h respectively. How long will they take to completely pass each other?

29. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the cars is:

30. Present ages of Sameer and Anand are in the ratio of 5 : 4 respectively. Three years hence, the ratio of their ages will become 11 : 9 respectively. What is Anand's present age in years?

31. Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11 : 10. What is Sagar's age at present?

32. A 20-liter mixture contains milk and water in the ratio 3:1. How much water must be added to make the ratio 1:1?

33. Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?

34. A milkman mixes water with milk. He has 60 liters of pure milk which cost him ₹50 per liter. He adds 20% water to the milk. If he sells the mixture at ₹55 per liter, find his total profit and profit percentage.

35. Pipes A and B can fill a tank in 5 and 6 hours respectively. Pipe C can empty it in 12 hours. If all the three pipes are opened together, then the tank will be filled in:

36. A train traveling at 60 kmph crosses another train traveling in the same direction at 24 kmph in 30 seconds. What is the combined length of both the trains?

37. Ratio of Speed of boat in still water and speed of stream is 7:2. Boat cover 120 km upstream in 8 hours and x km in 3 hours downstream. Find the value of x? (Hint: upstream = u – v km/hr, where u= speed of boat in still water, v= speed of the stream)

38. A boat can travel at speed of 13km/hr in still water. If the speed of the stream is 4km per hour, find the time taken by the boat to travel 51 km downstream? (Hint: downstream = u+v)

39. Two trains start from stations A and B spaced 50 kms apart at the same time and speed. As the trains start, a bird flies from one train towards the other and on reaching the second train, it flies back to the first train. This is repeated till the trains collide. If the speed of the trains is 25 km/h and that of the bird is 100km/h. How much did the bird travel till the collision. (Trains move in opposite direction)

40. A shopkeeper has 50 kg of rice. A part of which he sells at 10% profit and remaining at 5% loss. He gains 7% overall. Find the quantity sold at 10% profit.

41. A, B, and C are metals. A is 22 times as heavy as C and B is 3 times as heavy as C. In what ratio should A and B be mixed, so that the mixture is 13 times heavier than C?

42. Tea worth Rs. 126 per kg and Rs. 135 per kg are mixed with a third variety in the ratio 1 : 1 : 2. If the mixture is worth Rs. 153 per kg, the price of the third variety per kg will be:

43. In what ratio must a person mix three kinds of metal costing Rs.60/kg, Rs.75/kg and Rs.100 /kg so that the resultant alloy when sold at Rs.96/kg results in a gain of 20%?

44. There is a track with a length of 120 meters and 2 people, A & B, are running around it at 12 m/min and 20 m/min respectively in the same direction. When will B overtake A for the 1*st* time?

45. A can run one full round of a circular track in 6 min and B in 15 min. If both A and B start simultaneously from the same starting point then How many times would they met in the time B has completed 10 rounds when running in same direction, and In opposite direction?

46. A sum of money is to be distributed among A, B, C, D in the proportion of 5:2:4:3. If C gets ₹1000 more than D, what is B′s share?

47. At a carnival, the number of adult tickets sold at ₹120 each was 30 more than the number of child tickets sold at ₹85 each. What is the minimum number of each ticket sold if the total sales were at least ₹5400?

48. Write down numbers for the given place value expansion

a $5 \times 100 + 0 \times 10 + 0 \times 1 + 0 \times \frac{1}{10} + 0 \times \frac{1}{100} + 7 \times \frac{1}{1000}$

b $3 \times 100 + 2 \times 10 + 5 \times 1 + 6 \times \frac{1}{10} + 8 \times \frac{1}{100}$

c $1 \times 10 + 2 \times 1 + 3 \times \frac{1}{10}$

d $6 \times 1000 + 7 \times 100 + 8 \times 10 + 9 \times 1 + 1 \times \frac{1}{10} + 2 \times \frac{1}{100} + 3 \times \frac{1}{1000} + 4 \times \frac{1}{10000}$

49. Four runners are running on a circular track. They take 200 seconds, 300 seconds, 360 seconds, and 450 seconds respectively to complete one round. If they all start at the same point and at the same time, after how many minutes will they meet again at the starting point for the first time?

50. Four mechanical toys make a sound after every 3, 5, 6, and 9 minutes respectively. If they all make a sound together at 10:00 AM, how many times will they make a sound together between 10:00 AM and 1:00 PM (inclusive of the 10:00 AM sound)?

51. Find the HCF of 9/10, 12/25, 18/35, 21/40 ?

52. Express 0.456456456..... as a fraction in its simplest form.

53.

Using S_n to find a_n:

If the sum of the first n terms of a sequence is given by $S_n = n^2 + 5n$, find the nth term a_n and state if it's an arithmetic or geometric progression.

54. Find the sum of the series: 2+5+8+...+47.

55. The 3rd term of a geometric progression is 45, and the 6th term is 1215. Find the sum of the first 5 terms of this progression.

56. What is the sum of two consecutive even numbers, the difference of whose squares is 84?

57. A group of 10 friends planned to split the cost of a dinner equally. However, on the day of the dinner, 2 friends were unable to attend. The remaining friends still split the entire cost of the dinner equally. By what fraction did the share of each of the remaining friends increase?

58. A shopkeeper has 126 kg of rice and 198 kg of wheat. He wants to pack them into bags of equal size such that each bag contains either rice or wheat, and no food is left over. What is the maximum capacity of each bag?

59. Three different alarm clocks ring at intervals of 15 minutes, 20 minutes, and 30 minutes, respectively. If all three alarms rang together at 6:00 AM on Monday, June 9, 2025, at what exact time and date will they next ring together?

60. Two automated manufacturing machines, Machine X and Machine Y, are set to perform maintenance checks. Machine X performs a check every 15 minutes, and Machine Y performs a check every 25 minutes. If both machines started their checks at the same time, after how many minutes will they next perform a maintenance check simultaneously?

61. In a competitive exam, 8% of the applicants failed to meet the age criteria and were rejected. Among the accepted applicants, 60% opted for the engineering stream. If 10,800 accepted applicants chose other streams, then what was the total number of applicants for the competitive exam?

62. The numerator and denominator of a fraction are in the ratio 3:4. If 10 is added to the denominator, the result is a fraction that has a value 2:1 of the original fraction. The numerator of the original fraction is:

63. A bookstore sold a set of books at a final selling price after giving two successive discounts. The first discount was 20% on the marked price. The second discount was 5%. If the bookstore made a 14% profit on the cost price and the cost price of the set of books was ₹800, what was the initial marked price of the set of books?

64. In a 800-meter race, P starts 60 meters ahead of Q, but P ends up 40 meters behind Q when Q crosses the finish line. What is the ratio of the speeds of P to Q, assuming that both start at the same time?

65. A shopkeeper offers a discount of 10% on the marked price of an item. If he still makes a profit of 26%, what percentage above the cost price is the item marked?

66. Find the least number which when divided by 55, 66, 77, and 88 leaves a remainder 33, but when divided by 99 leaves no remainder.

1. A city has a total population of 15000, distributed among three districts: A, B, and C. Initially, the population ratio of District A to District B to District C is 5:3:2. Over a year, the following changes occur:
   - District A's population decreases by 10%.
   - District B's population increases by 15%.
   - District C's population changes such that the new ratio of District A's population to District C's population is 3:2.

After these changes, a survey reveals that 70% of District A's new population, 60% of District B's new population, and 75% of District C's new population are employed.
Question: What is the total number of unemployed people in the city after one year?

67. Ten years ago, a father's age was four times his son's age. Ten years from now, the father's age will be twice his son's age. What is the current age of the father and the son?

68. A runner covers a certain distance at an average speed. If the runner increases their speed by 2 km/hr, they complete the race 1 hour earlier. If the runner decreases their speed by 1 km/hr, they complete the race 1 hour later. What is the total distance of the race

69. A chemist has two solutions of acid.
    - Solution X contains 10% acid.
    - Solution Y contains 30% acid.

    How many liters of Solution X and how many liters of Solution Y should be mixed to obtain 15 liters of a solution containing 22% acid?

70. A person commutes to work by car. On Monday, they travel from home to office and back home, a total round trip distance.
    - On the way to the office, due to some traffic, their average speed was S km/hr.
    - On the way back home, the traffic cleared, and their average speed was 25% faster than their speed to the office.

    The total time taken for the entire round trip (home to office and office to home) was 2 hours and 45 minutes. If the one-way distance between home and office is 60 km, what was the average speed of the car on the way to the office?

71. A vendor purchases 100 shirts for his shop. He sells 60 of these shirts at a gain of 20% and the remaining 40 shirts at a gain of 10%. If he had sold all 100 shirts at a uniform profit of 18%, he would have received ₹50 more. Find the cost price of each shirt (in ₹).

72. In an examination A scored 20 marks more than B. If B has scored 5% less than A, how much has B scored?

73. A baker has a large number of cookies. When she tries to pack them into boxes containing exactly 12 cookies each, she finds that there are 5 cookies left over. If she tries to pack them into boxes containing exactly 15 cookies each, there are 8 cookies left over. And if she tries to pack them into boxes containing exactly 20 cookies each, there are 13 cookies left over.What is the smallest possible number of cookies the baker has?

74. Three runners are running on a circular track.The first runner completes one lap in 5/2 minutes. The second runner completes one lap in 7/4 minutes. The third runner completes one lap in 10/3 minutes. If they all start at the same point and at the same time, after how many minutes will they all meet again at the starting point for the first time?

75. A two-digit number is such that the sum of its digits is 9. If 27 is added to the number, its digits are reversed. Find the number.